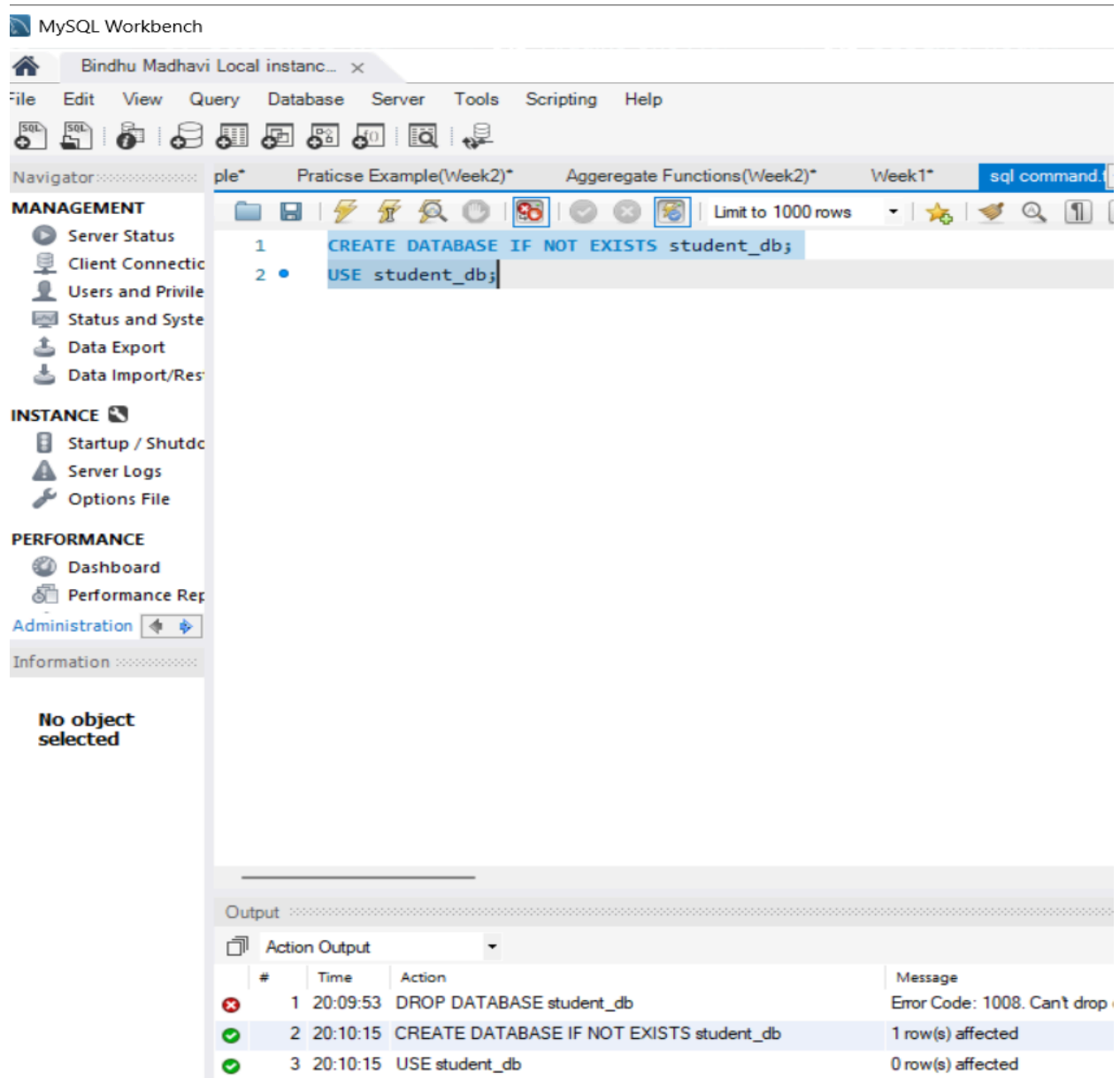


SQL COMMAND(WEEK1)

Name: P.Bindhu Madhavi

Roll Number: 2520030077

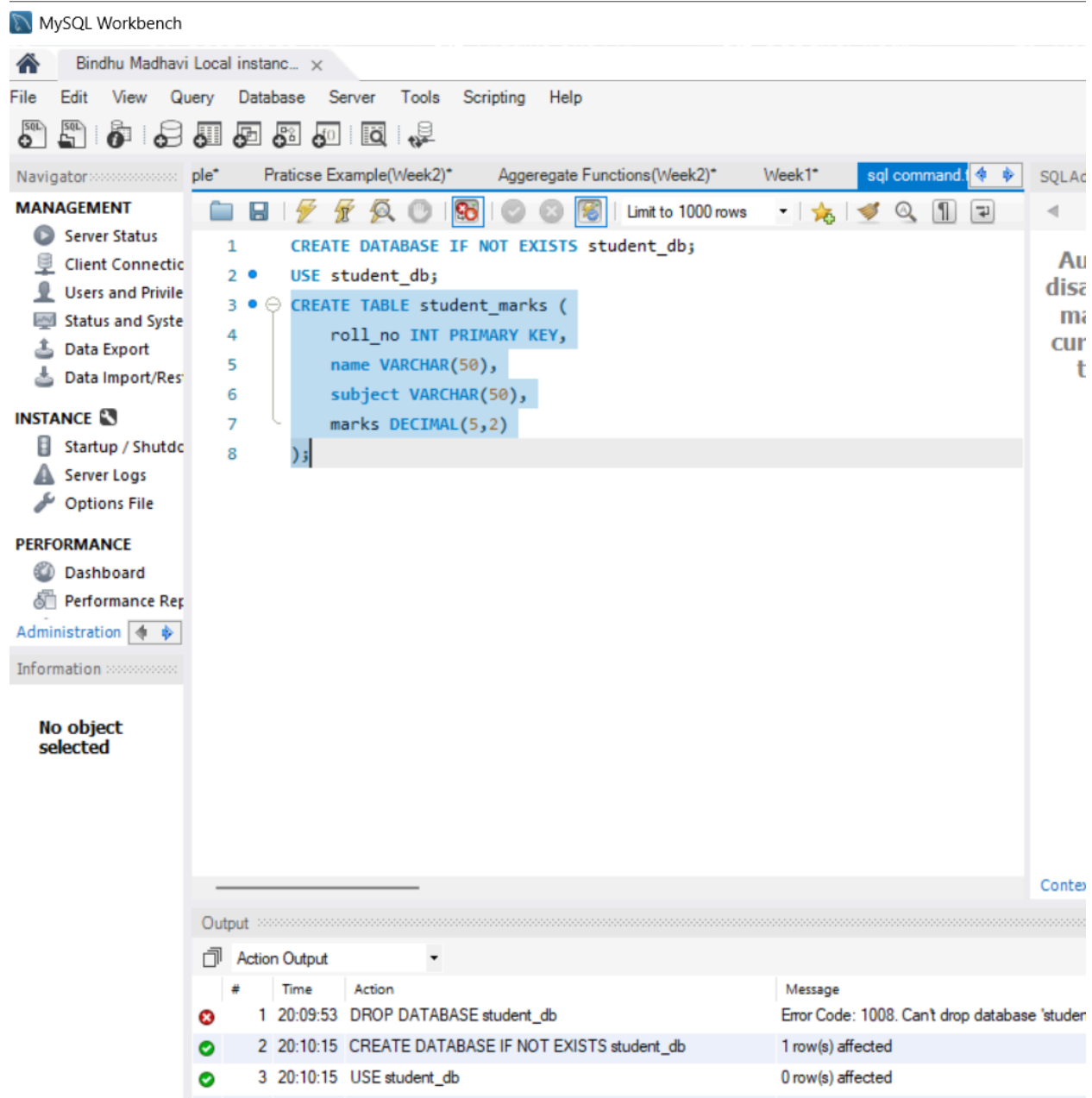
Step 1: Create Database



The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section has Dashboard and Performance Report. The 'Administration' section is also visible. The main window shows a SQL editor with two lines of code: `CREATE DATABASE IF NOT EXISTS student_db;` and `USE student_db;`. The output window at the bottom shows the execution results:

#	Time	Action	Message
1	20:09:53	DROP DATABASE student_db	Error Code: 1008. Can't drop
2	20:10:15	CREATE DATABASE IF NOT EXISTS student_db	1 row(s) affected
3	20:10:15	USE student_db	0 row(s) affected

Step 2: Create Table



The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section has Dashboard and Performance Report. The 'Administration' section is also visible. The main editor window displays the following SQL commands:

```
1 CREATE DATABASE IF NOT EXISTS student_db;
2 USE student_db;
3 CREATE TABLE student_marks (
4     roll_no INT PRIMARY KEY,
5     name VARCHAR(50),
6     subject VARCHAR(50),
7     marks DECIMAL(5,2)
8 );
```

The right sidebar shows the 'Autocomplete' section with suggestions for 'student_db' and 'student_marks'.

The bottom panel shows the 'Output' window with the following table:

#	Time	Action	Message
1	20:09:53	DROP DATABASE student_db	Error Code: 1008. Can't drop database 'student_db' (errno: 1008)
2	20:10:15	CREATE DATABASE IF NOT EXISTS student_db	1 row(s) affected
3	20:10:15	USE student_db	0 row(s) affected

Step 3: Insert Records

MySQL Workbench

Bindhu Madhavi Local instanc...

File Edit View Query Database Server Tools Scripting Help

Navigator: ...ple* Praticse Example(Week2)* Aggreagate Functions(Week2)* Week1* sql command

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

Result Grid

	roll_no	name	subject	marks
1	Bindhu Madhavi	Math	85.50	
2	Bindhu Madhavi	Math	92.75	
3	Anil	Math	78.40	
4	Priya	Math	88.90	
5	Vijay	Math	80.25	
6	subbusir	aws	98.50	
7	dwaraka	dbms	95.75	
8	ranjani	english	97.40	
9	kavithamam	aws1	99.90	
10	seetha	azure	82.25	
*	NULL	NULL	NULL	

student_marks 1 x

Output

Action Output

#	Time	Action	Message
3	20:10:15	USE student_db	0 row(s) affected
4	20:14:22	CREATE TABLE student_marks (roll_no INT PRIMAR...	0 row(s) affected
5	20:15:27	INSERT INTO student_marks (roll_no, name, subject, mar...	10 row(s) affected Records: 10
6	20:16:22	SELECT * FROM student_marks LIMIT 0, 1000	10 row(s) returned

Object Info

Step 5: Aggregate Functions

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: Bindhu Madhavi Local instance... x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

Pratise Example(Week2)* Aggregate Functions(Week2)* Week1* sql command

Limit to 1000 rows

```

19 (10, 'seetha', 'azure', 82.25);
20 SELECT * FROM student_marks;
21 SELECT COUNT(*) AS total_students FROM student_marks;
22 SELECT SUM(marks) FROM student_marks;
23 SELECT AVG(marks) FROM student_marks;
24 SELECT MAX(marks) FROM student_marks;
25 SELECT MIN(marks) FROM student_marks;

```

Result Grid

MIN(marks)
78.40

Export: Wrap Cell Content:

student_marks 2 Result 3 Result 4 Result 5 Result 6 Result 7 Read Only

Output

Action Output

#	Time	Action	Message
9	20:16:53	SELECT SUM(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned
10	20:16:53	SELECT AVG(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned
11	20:16:53	SELECT MAX(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned
12	20:16:54	SELECT MIN(marks) FROM student_marks LIMIT 0, 1000	1 row(s) returned

Step 6: Where Clause

MySQL Workbench interface showing a query session for 'Bindhu Madhavi Local instance...'.

Query Editor: The SQL editor contains the following queries:

```

27 • student_marks WHERE marks > 85;
28 • student_marks WHERE marks >= 90;
29 • student_marks WHERE marks < 80;
30 • student_marks WHERE marks BETWEEN 80 AND 90;
31 • student_marks WHERE name LIKE 'P%';
32 • student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
33 • student_marks WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'P%');

```

Result Grid: The result grid shows the following data:

roll_no	name	subject	marks
1	Bindhu Madhavi	Math	85.50
2	Bindhu Madhavi	Math	92.75
4	Priya	Math	88.90
* NULL	NULL	NULL	NULL

Output: The output pane shows the execution results of the queries:

#	Time	Action	Message
✓ 22	20:17:48	SELECT * FROM student_marks WHERE marks BETWEEN...	4 row(s) returned
✓ 23	20:17:48	SELECT * FROM student_marks WHERE name LIKE 'P%...	1 row(s) returned
✓ 24	20:17:48	SELECT * FROM student_marks WHERE name IN ('Ravi'...	1 row(s) returned
✓ 25	20:17:48	SELECT * FROM student_marks WHERE marks > 85 AN...	3 row(s) returned

Step 7: Update Commands

The screenshot shows the MySQL Workbench interface with the following components:

- Navigator:** Displays the database structure, including tables like `student_marks`.
- SQL Editor:** Contains the following SQL commands:


```

14 (5, 'Vijay', 'Math', 80.25),
15 (6, 'subbusir', 'aws', 98.50),
16 (7, 'dwaraka', 'dbms', 95.75),
17 (8, 'ranjani', 'english', 97.40),
18 (9, 'kavithamam', 'aws1', 99.90),
19 (10, 'seetha', 'azure', 82.25);
20 • SELECT * FROM student_marks;
21 • SELECT COUNT(*) AS total_students FROM student_marks;
22 • SELECT SUM(marks) FROM student_marks;
23 • SELECT AVG(marks) FROM student_marks;
24 • SELECT MAX(marks) FROM student_marks;
25 • SELECT MIN(marks) FROM student_marks;
26
27 • SELECT * FROM student_marks WHERE marks > 85;
28 • SELECT * FROM student_marks WHERE marks >= 90;
29 • SELECT * FROM student_marks WHERE marks < 80;
30 • SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
31 • SELECT * FROM student_marks WHERE name LIKE 'P%';
32 • SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
33 • SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR na
34 • SET SQL_SAFE_UPDATES = 0;
      
```
- Output:** Shows the execution results of the commands:

#	Time	Action	Message
✓ 23	20:17:48	SELECT * FROM student_marks WHERE name LIKE 'P%...	1 row(s) returned
✓ 24	20:17:48	SELECT * FROM student_marks WHERE name IN ('Ravi'...	1 row(s) returned
✓ 25	20:17:48	SELECT * FROM student_marks WHERE marks > 85 AN...	3 row(s) returned
✓ 26	20:18:40	SET SQL_SAFE_UPDATES = 0	0 row(s) affected

Step 8: Update Commands

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: Bindhu Madhavi Local instanc... x

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

Practicse Example(Week2)*

```

16 (7, 'dwaraka', 'dbms', 95.75),
17 (8, 'ranjani', 'english', 97.40),
18 (9, 'kavithamam', 'aws1', 99.90),
19 (10, 'seetha', 'azure', 82.25);
20 • SELECT * FROM student_marks;
21 • SELECT COUNT(*) AS total_students FROM student_marks;
22 • SELECT SUM(marks) FROM student_marks;
23 • SELECT AVG(marks) FROM student_marks;
24 • SELECT MAX(marks) FROM student_marks;
25 • SELECT MIN(marks) FROM student_marks;
26
27 • SELECT * FROM student_marks WHERE marks > 85;
28 • SELECT * FROM student_marks WHERE marks >= 90;
29 • SELECT * FROM student_marks WHERE marks < 80;
30 • SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
31 • SELECT * FROM student_marks WHERE name LIKE 'P%';
32 • SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
33 • SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR na
34 • SET SQL_SAFE_UPDATES = 0;
35 • UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;
36 • UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80;

```

Output

Action Output

#	Time	Action	Message
✓ 25	20:17:48	SELECT * FROM student_marks WHERE marks > 85 AN...	3 row(s) returned
✓ 26	20:18:40	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
✓ 27	20:19:11	UPDATE student_marks SET marks = 90.00 WHERE roll...	1 row(s) affected Rows matched: 1 Char
✓ 28	20:19:11	UPDATE student marks SET subject = 'Remedial Math' ...	0 row(s) affected Rows matched: 0 Char

Step 9: Delete Commands

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: ple* Praticse Example(Week2)* Aggreagate Functions(Week2)* Week1* sql command SQL

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdo
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```
21 • SELECT COUNT(*) AS total_students FROM student_marks;
22 • SELECT SUM(marks) FROM student_marks;
23 • SELECT AVG(marks) FROM student_marks;
24 • SELECT MAX(marks) FROM student_marks;
25 • SELECT MIN(marks) FROM student_marks;
26
27 • SELECT * FROM student_marks WHERE marks > 85;
28 • SELECT * FROM student_marks WHERE marks >= 90;
29 • SELECT * FROM student_marks WHERE marks < 80;
30 • SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
31 • SELECT * FROM student_marks WHERE name LIKE 'P%';
32 • SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
33 • SELECT * FROM student_marks WHERE marks > 85 AND (subject = 'Math' OR na
34 • SET SQL_SAFE_UPDATES = 0;
35 • UPDATE student_marks SET marks = 90.00 WHERE roll_no = 3;
36 • UPDATE student_marks SET subject = 'Remedial Math' WHERE marks < 80;
37
38 • DELETE FROM student_marks WHERE roll_no = 5;
39 • SET SQL_SAFE_UPDATES = 0;
40 • DELETE FROM student_marks WHERE marks < 75;
41 • SET SQL_SAFE_UPDATES = 1;
```

Output

Action Output

#	Time	Action	Message
✓ 29	20:19:41	DELETE FROM student_marks WHERE roll_no = 5	1 row(s) affected
✓ 30	20:19:41	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
✓ 31	20:19:41	DELETE FROM student_marks WHERE marks < 75	0 row(s) affected
✓ 32	20:19:41	SET SQL_SAFE_UPDATES = 1	0 row(s) affected

Object Info

Step 10: Order By Clause

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigators Bindhu Madhavi Local instanc... x

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

No object selected

practise* Praticse Example(Week2)* Aggreagate Functions(Week2)* Week1* sql command.

Limit to 1000 rows

```
26
27 • SELECT * FROM student_marks WHERE marks > 85;
28 • SELECT * FROM student_marks WHERE marks >= 90;
29 • SELECT * FROM student_marks WHERE marks < 80;
30 • SELECT * FROM student_marks WHERE marks BETWEEN 80 AND 90;
31 • SELECT * FROM student_marks WHERE name LIKE 'P%';
32 • SELECT * FROM student_marks WHERE name IN ('Ravi', 'Sita', 'Vijay');
```

Result Grid

	roll_no	name	subject	marks
▶	6	subbusir	aws	98.50
	10	seetha	azure	82.25
	8	ranjani	english	97.40
	4	Priya	Math	88.90
	9	kavithamam	aws1	99.90
	7	dwaraka	dbms	95.75
	1	Bindhu Madhavi	Math	85.50
	2	Bindhu Madhavi	Math	92.75
	3	Anil	Math	90.00
*	NULL	NULL	NULL	NULL

marks 33 student_marks 34 student_marks 35 student_marks 36 stuc Apply R

Output

Action Output

#	Time	Action	Message
✓ 57	20:20:23	SELECT * FROM student_marks ORDER BY marks ASC ...	9 row(s) returned
✓ 58	20:20:23	SELECT * FROM student_marks ORDER BY marks DES...	9 row(s) returned
✓ 59	20:20:23	SELECT * FROM student_marks ORDER BY name ASC ...	9 row(s) returned
✓ 60	20:20:23	SELECT * FROM student_marks ORDER BY name DESC...	9 row(s) returned

Object Info

Step 11: Calculate Total Marks Subject-wise (Group By Clause)

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: Bindhu Madhavi Local instanc... x

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

Praticse Example(Week2)* Aggreagate Functions(Week2)* Week1* sql command

```
43
44 • SELECT * FROM student_marks ORDER BY marks ASC;
45 • SELECT * FROM student_marks ORDER BY marks DESC;
46 • SELECT * FROM student_marks ORDER BY name ASC;
47 • SELECT * FROM student_marks ORDER BY name DESC;
48 • SELECT subject, SUM(marks) AS total_marks FROM
49 student_marks GROUP BY subject;
```

Result Grid

	subject	total_marks
▶	Math	357.15
	aws	98.50
	dbms	95.75
	english	97.40
	aws1	99.90
	azure	82.25

Result 38 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 58	20:20:23	SELECT * FROM student_marks ORDER BY marks DES...	9 row(s) returned
✓ 59	20:20:23	SELECT * FROM student_marks ORDER BY name ASC ...	9 row(s) returned
✓ 60	20:20:23	SELECT * FROM student_marks ORDER BY name DESC...	9 row(s) returned
✓ 61	20:23:37	SELECT subject, SUM(marks) AS total_marks FROM stud...	6 row(s) returned

Object Info

Step 12: Calculate Average Marks Subject-wise (Group By Clause)

MySQL Workbench

Bindhu Madhavi Local instanc... x

File Edit View Query Database Server Tools Scripting Help

Navigator: Bindhu Madhavi Local instance

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Report

Administration

Information

Pratice Example(Week2)* Aggregate Functions(Week2)* Week1* sql command

```
45 • SELECT * FROM student_marks ORDER BY marks DESC;
46 • SELECT * FROM student_marks ORDER BY name ASC;
47 • SELECT * FROM student_marks ORDER BY name DESC;
48 • SELECT subject, SUM(marks) AS total_marks FROM
49 student_marks GROUP BY subject;
50 • SELECT subject, AVG(marks) AS avg_marks FROM
51 student_marks GROUP BY subject;
```

Result Grid

	subject	avg_marks
▶	Math	89.287500
	aws	98.500000
	dbms	95.750000
	english	97.400000
	aws1	99.900000
	azure	82.250000

Result 39 x Read Only

Output

Action Output

#	Time	Action	Message
✓ 59	20:20:23	SELECT * FROM student_marks ORDER BY name ASC ...	9 row(s) returned
✓ 60	20:20:23	SELECT * FROM student_marks ORDER BY name DESC...	9 row(s) returned
✓ 61	20:23:37	SELECT subject, SUM(marks) AS total_marks FROM stud...	6 row(s) returned
✓ 62	20:24:08	SELECT subject, AVG(marks) AS avg_marks FROM stud...	6 row(s) returned

Step 13: Count Students Subject-wise(Group By Clause)

MySQL Workbench

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: ...ple* Practise Example(Week2)* Aggreagate Functions(Week2)* Week1* sql command. SQLAdditions

MANAGEMENT

- Server Status
- Client Connectio
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration Information

No object selected

```

47 • SELECT * FROM student_marks ORDER BY name DESC;
48 • SELECT subject, SUM(marks) AS total_marks FROM
49 student_marks GROUP BY subject;
50 • SELECT subject, AVG(marks) AS avg_marks FROM
51 student_marks GROUP BY subject;
52 • SELECT subject, COUNT(*) AS num_students FROM
53 student_marks GROUP BY subject;

```

Limit to 1000 rows

Result Grid

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

Exports: Wrap Cell Content: **Result Grid** Form Editor Field Types Query Stats

Result 40 Read Only Context Help

Output

Action Output

#	Time	Action	Message
60	20:20:23	SELECT * FROM student_marks ORDER BY name DESC...	9 row(s) returned
61	20:23:37	SELECT subject, SUM(marks) AS total_marks FROM stud...	6 row(s) returned
62	20:24:08	SELECT subject, AVG(marks) AS avg_marks FROM stud...	6 row(s) returned
63	20:24:49	SELECT subject, COUNT(*) AS num_students FROM stu...	6 row(s) returned

Step 14: Display Subjects with Average Marks Greater Than 90 (Having Clause)

MySQL Workbench interface showing a query execution session.

Query Editor:

```

52 • SELECT subject, COUNT(*) AS num_students FROM
53 student_marks GROUP BY subject;
54 • SELECT subject, AVG(marks) AS avg_marks
55 FROM student_marks
56 GROUP BY subject
57 HAVING AVG(marks) > 90;
58

```

Result Grid:

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

Output:

#	Time	Action	Message
61	20:23:37	SELECT subject, SUM(marks) AS total_marks FROM stud...	6 row(s) returned
62	20:24:08	SELECT subject, AVG(marks) AS avg_marks FROM stud...	6 row(s) returned
63	20:24:49	SELECT subject, COUNT(*) AS num_students FROM stu...	6 row(s) returned
64	20:25:17	SELECT subject, AVG(marks) AS avg_marks FROM stud...	4 row(s) returned

Step 15: Display Subjects Having More Than One Student (Having Clause)

Bindhu Madhavi Local instance...

File Edit View Query Database Server Tools Scripting Help

Navigator: ... ple* Praticse Example(Week2)* Aggreagate Functions(Week2)* Week1* sql command. SQLAddit

MANAGEMENT

- Server Status
- Client Connectic
- Users and Privile
- Status and Syste
- Data Export
- Data Import/Res

INSTANCE

- Startup / Shutdc
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Rep

Administration

Information

No object selected

```
55 FROM student_marks
56 GROUP BY subject
57 HAVING AVG(marks) > 90;
58 • SELECT subject, COUNT(*) AS num_students
59 FROM student_marks
60 GROUP BY subject
61 HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	subject	num_students
▶	Math	4

Result 42 x Read Only Context H

Output

Action Output

#	Time	Action	Message
✓ 62	20:24:08	SELECT subject, AVG(marks) AS avg_marks FROM stud...	6 row(s) returned
✓ 63	20:24:49	SELECT subject, COUNT(*) AS num_students FROM stu...	6 row(s) returned
✓ 64	20:25:17	SELECT subject, AVG(marks) AS avg_marks FROM stud...	4 row(s) returned
✓ 65	20:25:51	SELECT subject, COUNT(*) AS num_students FROM stu...	1 row(s) returned